

Stochastic Nonconvex Optimization with Heavy-Tailed Noise

Distance and Suboptimality Dependence

Zhankun Luo
MINDS Group
Purdue University

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Machine
Intelligence
& Networked
Data
Science

Project motivation

A uniform bound on gradient noise assigns the same moment budget to every point in the domain.

Noise can vary across the landscape

The permitted noise level may depend on how far the algorithm moves, or on how far the objective value is from its infimum.

Heavy tails add a statistical difficulty

We allow a finite p -th noise moment with $1 < p \leq 2$. When $p < 2$, the variance can be infinite.

How do tail behavior and noise growth affect the number of stochastic gradient evaluations?

Background: [Fazla et al. '26; Fradin et al. '26].

Optimization problem and oracle

We consider an unconstrained, smooth nonconvex objective:

$$\min_{\mathbf{x} \in \mathbb{R}^d} f(\mathbf{x}), \quad f_* := \inf_{\mathbf{x} \in \mathbb{R}^d} f(\mathbf{x}) > -\infty,$$

with a fixed initialization $\mathbf{x}_0$ and $f(\mathbf{x}_0) - f_* \leq \Delta$.

Unbiased stochastic information

$$\mathbf{h}(\mathbf{x}; \xi) = \nabla f(\mathbf{x}) + \zeta(\mathbf{x}; \xi), \quad \mathbb{E}_{\xi}[\zeta(\mathbf{x}; \xi)] = \mathbf{0}.$$

Accuracy and cost

We seek a random output $\mathbf{X}$ with $\mathbb{E}\|\nabla f(\mathbf{X})\| \leq \epsilon$. Count every returned stochastic vector, including each sample inside a batch.

Noise moments and growth profiles

At every $\mathbf{x} \in \mathbb{R}^d$, we impose a centered moment bound

$$\mathbb{E}_{\xi} \|\zeta(\mathbf{x}; \xi)\|^p \leq M(\mathbf{x}), \quad 1 < p \leq 2.$$

Moment order p

- $p = 2$: finite variance at each point.
- $1 < p < 2$: finite p -th moment, with possibly infinite variance.

Moment budget $M(\mathbf{x})$

- The budget may vary with the query point.
- It bounds the moment from above.
- The actual moment need not attain the budget.

A general growth profile q will describe how the budget varies.

Distance and suboptimality models

Let $B, G \geq 0$ and let $q : [0, \infty) \rightarrow [0, \infty)$ be finite and nondecreasing.

Distance model: tentative name q-BG

$$\mathbb{E}_{\xi} \|\zeta(\mathbf{x}; \xi)\|^p \leq Bq(\|\mathbf{x} - \mathbf{x}_0\|) + G.$$

The budget depends on distance from the fixed initialization.

Suboptimality model: tentative name q-ABC

$$\mathbb{E}_{\xi} \|\zeta(\mathbf{x}; \xi)\|^p \leq Bq(f(\mathbf{x}) - f_*) + G.$$

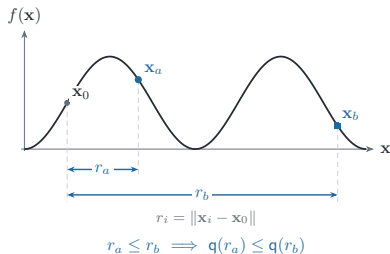
The budget depends on the objective gap.

B scales the growth term and G is the additive baseline. We study these as two alternative oracle models.

Distance and suboptimality in the landscape

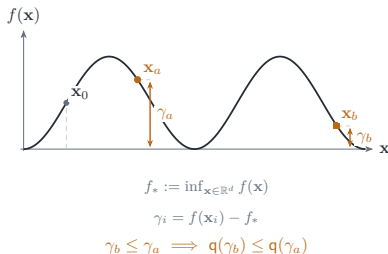
(a) Distance-dependent noise: q-BG

$$\mathbb{E}_{\xi} \|\zeta(\mathbf{x}; \xi)\|^p \leq Bq(\|\mathbf{x} - \mathbf{x}_0\|) + G$$



(b) Gap-dependent noise: q-ABC

$$\mathbb{E}_{\xi} \|\zeta(\mathbf{x}; \xi)\|^p \leq Bq(f(\mathbf{x}) - f_*) + G$$



Nondecreasing q orders the permitted moment budgets.

Along this slice, $\mathbf{x}_b$ is farther from $\mathbf{x}_0$ than $\mathbf{x}_a$, but has a smaller gap. For nondecreasing q , its distance-model budget is weakly larger and its gap-model budget is weakly smaller.



Ordinary and mean-square smoothness

Ordinary smoothness

The mean objective has an L -Lipschitz gradient:

$$\|\nabla f(\mathbf{x}) - \nabla f(\mathbf{y})\| \leq L\|\mathbf{x} - \mathbf{y}\|.$$

The baseline oracle returns one vector using a fresh sample per call.

Mean-square smoothness (MSS)

A stronger condition controls oracle differences:

$$\mathbb{E}_{\xi} \|\mathbf{h}(\mathbf{x}; \xi) - \mathbf{h}(\mathbf{y}; \xi)\|^2 \leq \bar{L}^2 \|\mathbf{x} - \mathbf{y}\|^2.$$

Both terms use the **same sample ξ** . For the MSS setting, specify whether paired evaluations are available and count a pair as two vector evaluations.

Research questions

The project studies both noise models under both smoothness regimes.

Noise model	Ordinary smoothness	MSS
Distance: q-BG	Lower & upper bounds	Lower & upper bounds
Gap: q-ABC	Lower & upper bounds	Lower & upper bounds

- How do the moment order p and growth profile q interact?
- Which profiles allow matching bounds?
- How do smoothness and access to paired samples change the oracle cost?

Initial profiles of interest include $q(s) = s^a$, $a > 0$.

Planned lower bounds and algorithms

Lower-bound direction

- Construct hard objectives in regions with large noise budgets.
- Use oracles that reveal little information per call.
- Check moment bounds over the entire domain.

Goal: compare lower bounds and algorithmic cost under the same assumptions.

Starting points: [Fazla et al. '26; Fradin et al. '26; Zhang et al. '20].

Algorithmic direction

- Explore normalization to control movement.
- Adapt batch sizes or step sizes to local noise budgets.
- Study variance reduction with paired MSS samples.

References

- Fazla, A., E. C. Kaya, A. Upadhyay, and A. Hashemi (2026). *Lower Bounds and Proximally Anchored SGD for Non-Convex Minimization Under Unbounded Variance*. [arXiv: 2604.16620](#). [Paper](#).
- Fradin, A., A. Sadiev, L. Condat, and P. Richtárik (2026). *Tight Lower Bounds and Optimal Algorithms for Stochastic Nonconvex Optimization with Heavy-Tailed Noise*. [arXiv: 2512.18713](#). [Paper](#).
- Zhang, J., S. P. Karimireddy, A. Veit, S. Kim, S. Reddi, S. Kumar, and S. Sra (2020). *Why are Adaptive Methods Good for Attention Models?* [arXiv: 1912.03194](#). [Paper](#).